Hybrid Approaches to Multi-Agent Path Finding: How Environment Shapes Algorithm Performance

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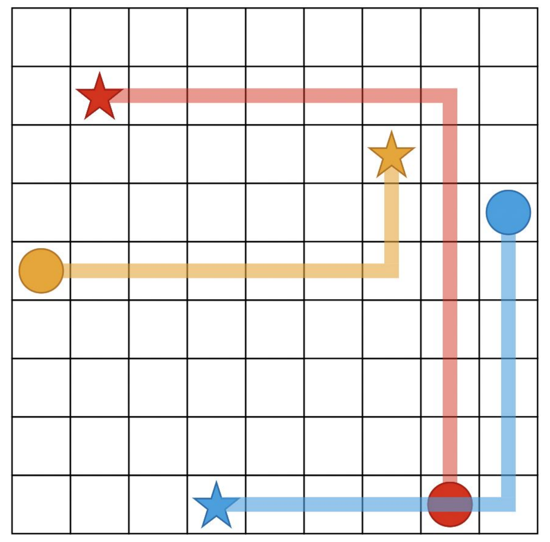
BACKGROUND

Multi-Agent Path Finding (MAPF) studies how to compute **collision-free paths** for multiple robots moving in a shared environment. Each agent must reach its goal while avoiding collisions with others, planning in both space and time.



Why is this challenging?

- Exponential growth of the joint search space
- Interdependent collision constraints and cascading conflicts
- Tradeoff between optimality and scalability

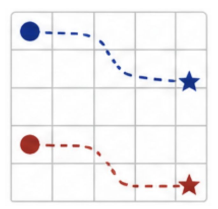
MAPF is a scalability problem ⟹ as environments and agent interactions grow more complex, efficient coordination becomes increasingly difficult.

APPROACHES

MAPF approaches differ in how they coordinate agents, leading to fundamental tradeoffs between scalability and solution quality.

Decoupled

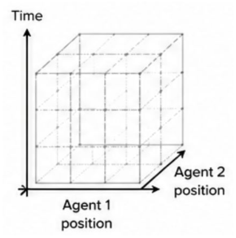
Plan agents independently



- Fast, scalable
- Not complete, can fail in dense settings

Coupled

Plans all robots at same time in the joint space



- Complete, optimal
- Exponential complexity, very slow

Hybrid

Plan independently, resolve conflicts jointly

- Balances scalability and optimality
- Performance depends on environment

Hybrid algorithms give high-quality solutions without the exponential cost of full joint planning

Hybrid MAPF Algorithms Compared

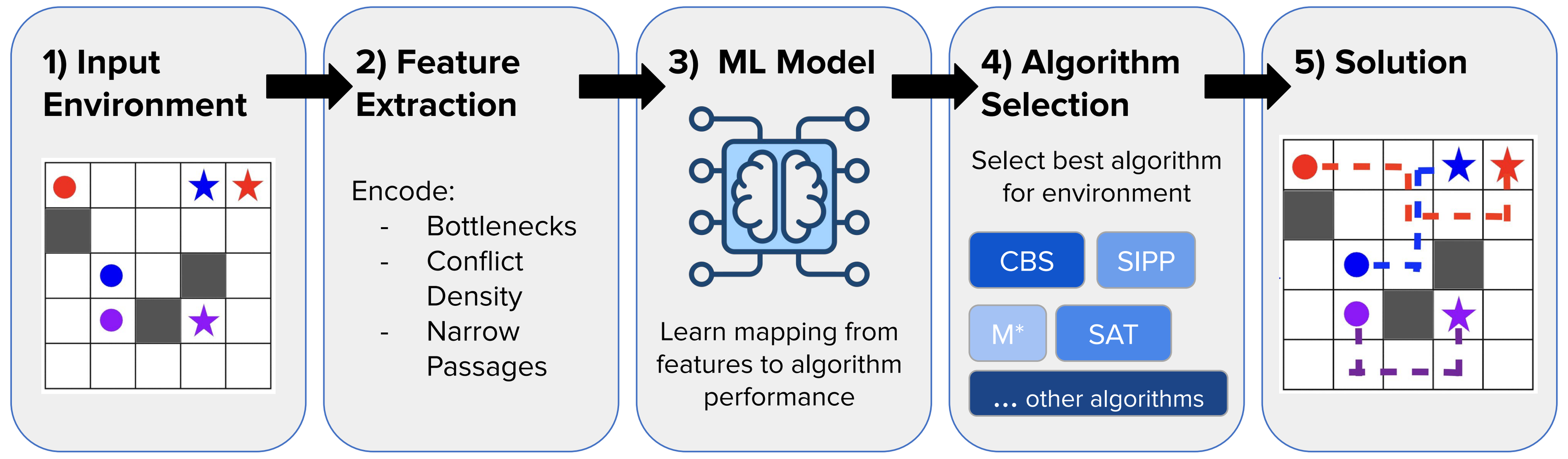
Algorithm	Hybrid Functionality	Works Best When
CBS	Resolve conflicts using a constraint tree	Few conflicts, structured environments
M*	Couple agents only when conflicts occur	Sparse, localized interactions
SIPP	Plan using safe time intervals	Timing constraints, dynamic settings
Incremental Coordination	Plan sequentially and adjust for conflicts	Moderate conflict density
ICBS / ICobot	Improved CBS with heuristics	Larger, more complex CBS instances
SAT Solver	Encode MAPF as a global constraint problem	Dense, highly constrained environments

No hybrid algorithm dominates across all environments - conflict density, topology, and interaction patterns affect which strategy wins. This motivates our adaptive framework.

OUR RESEARCH

Our Goal: Build an adaptive framework that understands the environment and selects the most appropriate MAPF strategy to efficiently find high-quality solutions.

Algorithm Selection Pipeline



Research Goals

1) Better Problem Understanding	2) Better Decoupling & Adaptive Planning	3) MAPF For Advanced Planning
Learn to **characterize** MAPF **environments** so the right algorithm can be selected before planning begins. • Identify and encode key structural features: **bottlenecks, conflict density, narrow passages** • Use extracted features to **predict** and guide **algorithm selection**	Reduce planning **complexity** without sacrificing solution quality by coordinating agents more intelligently. • Avoid planning for all agents simultaneously - **partition** agents by **interaction locality** • Develop effective **scheduling strategies** that balance independence and coordination	Extend MAPF beyond idealized grids to guide planners in more **realistic, physically constrained** settings. • Incorporate real-world kinematic constraints: **acceleration, velocity, turning radius** • Use MAPF as a **coordination layer** for higher-level, complex planners

METHODS

FRAMEWORK AND IMPLEMENTATION

Python Framework Selection

Chose an open-source Python MAPF library for its:
- rich built-in environments
- strong visualization tools
- existing implementations of CBS and SIPP

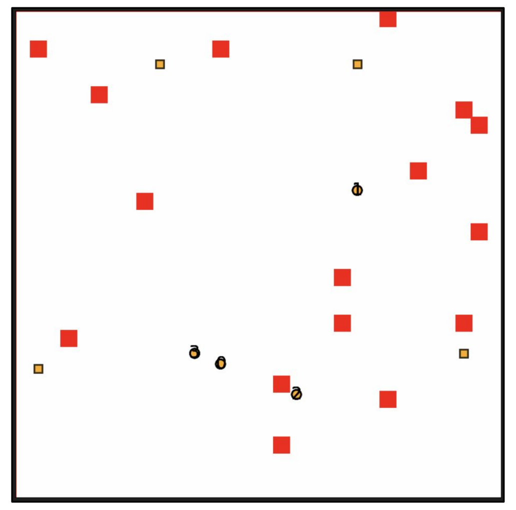
This library gives us a reliable baseline to extend.

Algorithms Implemented

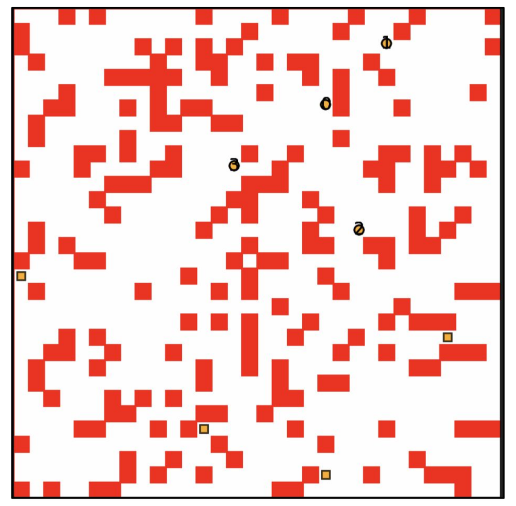
Extended the framework with additional hybrid solvers to enable a full comparative study across environment types.

EXPERIMENTAL DESIGN

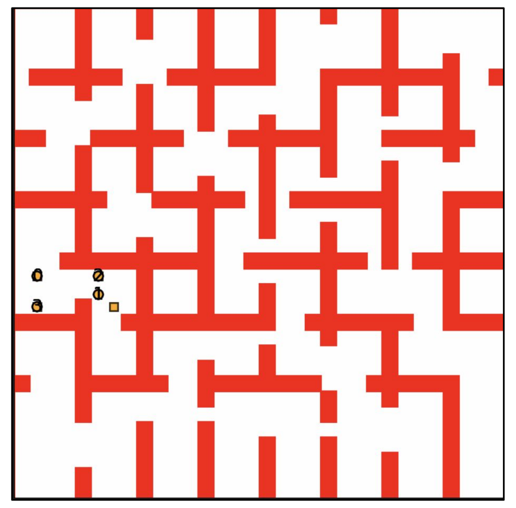
Open Environment



Cluttered Environment



Narrow Passage Environment



TESTING SCALABILITY AND EFFECTS OF ENVIRONMENT

To systematically evaluate each algorithm, we have constructed a benchmarking pipeline that runs every solver across all three environment types (open, cluttered, and narrow passage) at seven agent scales: 2, 4, 8, 16, 32, 64, and 128 robots. For each run, the pipeline automatically records four performance metrics:

- **Runtime** - total wall-clock planning time
- **Solution cost** - sum of all agent path lengths
- **Makespan** - time steps until the last agent reaches its goal
- **Scalability** - how each metric degrades as agent count doubles

Data collection is currently underway. Once complete, results will be analyzed to identify which algorithms perform best under each environment type and agent load — directly informing the training data for our adaptive ML selectors

2 → 4 → 8 → 16 → 32 → 64 → 128 Robots

FUTURE WORK

Having established a benchmarking pipeline across environments and agent scales, our next step is to analyze *which environmental characteristics most strongly predict algorithm performance* - identifying whether factors like obstacle density, passage width, or agent interaction locality are the dominant drivers of each solver's behavior.

From there, we will develop principled methods to **quantify and encode these features** as numerical descriptors, transforming raw environment structure into a feature vector that captures what matters for algorithm selection.

The ultimate goal is a fully adaptive MAPF framework that selects its strategy automatically, generalizing across the wide range of problem instances encountered in real-world multi-robot circumstances.

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